Model Predictive Control — Spring 2026
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Solution 06

Exercise 1

Let $\mathcal{X} \subset \mathbb{R}^n$ be a polytope with vertices v_i, $i = 1, 2, .., m$. Show that $\mathcal{X}$ is an invariant set for the system $x(k+1) = Ax(k)$ if and only if it holds for all $i = 1, 2, .., m$ that $Av_i \in \mathcal{X}$.

Hint 1: For a polytope $\mathcal{X} \subset \mathbb{R}^n$ every $x \in \mathcal{X}$ can be written as a convex combination of the vertices $x = \sum_{i=1}^m \lambda_i v_i$, $\sum_{i=1}^m \lambda_i = 1$, $\lambda_i \geq 0$.

Hint 2: Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a convex set and consider vectors $y^1, y^2, .., y^k \in \mathcal{X}$. It holds that the convex hull $co(y^1, y^2, .., y^k) \subseteq \mathcal{X}$.

Solution:

What we want to show: $\mathcal{X}$ invariant for $x(k+1) = Ax(k) \Leftrightarrow \forall$ vertices v_i of $\mathcal{X}$, it holds that $Av_i \in \mathcal{X}$.

Recall the invariance definition (Chap. 5, Slide 21): A set $\mathcal{X}$ is said to be a positive invariant set for the autonomous system $x(k+1) = Ax(k)$ if for all $x(k) \in \mathcal{X}$ it holds that $x(k+1) = Ax(k) \in \mathcal{X}$.

Necessity ($\Rightarrow$):

- All vertices v_i are included in the polytope $\mathcal{X}$, i.e., $v_i \in \mathcal{X}$.
- By the invariance assumption of $\mathcal{X}$, it follows that $Av_i \in \mathcal{X}$.

Sufficiency ($\Leftarrow$):

- Let $x \in \mathcal{X}$.
- For the next time step we know from Hint 1 that there exists λ_i such that

$$Ax = A\sum_{i=1}^m \lambda_i v_i = \sum_{i=1}^m \lambda_i \underbrace{Av_i}_{\tilde{v}_i} = \sum_{i=1}^m \lambda_i \tilde{v}_i.$$

  with $\sum_{i=0}^m \lambda_i = 1$ and $\lambda_i \geq 0$ for all $i = 1, \ldots, m$.
- By assumption it holds $Av_i \in \mathcal{X}$, implying that $\tilde{v}_i \in \mathcal{X}$.
- It follows that Ax is in the convex hull of the points $\tilde{v}_i$, i.e.,

$$Ax = \sum_{i=1}^m \lambda_i \tilde{v}_i \in \text{co}(\tilde{v}_1, \ldots, \tilde{v}_m).$$

- From Hint 2 we know that since $\mathcal{X}$ convex and $\tilde{v}_i \in \mathcal{X}$, it holds that the convex hull $\text{co}(\tilde{v}_1, \ldots, \tilde{v}_m) \subseteq \mathcal{X}$.
- We can conclude invariance of $\mathcal{X}$ because $x(k+1) = Ax(k) \in \mathcal{X} \quad \forall x \in \mathcal{X}$

Exercise 2

Consider the MPC problem

$$
\begin{aligned}
&\min_U \; p x_N^2 + \sum_{i=0}^{N-1} q x_i^2 + r u_i^2 \\
&\text{s.t. } x_{i+1} = 1.5 x_i + u_i \\
&\qquad -0.1 \leq u_i \leq 0.1 \\
&\qquad x_0 = x(k) \\
&\qquad x_N \in \mathcal{X}_f,
\end{aligned}
$$

where $U = [u_0, \ldots, u_{N-1}]^T$, $q \geq 0$, $p > 0$, and $r > 0$. What is the largest terminal invariant set $\mathcal{X}_f$ for the problem under a stabilizing linear terminal control law, i.e., $u(k) = Kx(k)$? (For this exercise, Lyapunov stability is sufficient for a controller to be stabilizing, that is, we don't require it to result in an asymptotically stable closed loop.)

1. $[-0.15,\ 0.15]$ 2. $[-1.5,\ 1.5]$ 3. $[-0.2,\ 0.2]$ 4. $[-0.5,\ 0.5]$

Hint: Algorithm to compute maximum positively invariant set Ω_∞:

1. init $\Omega_0 \leftarrow \mathcal{X}$
2. $\Omega_{i+1} \leftarrow \text{pre}(\Omega_i) \cap \Omega_i$
3. if $\Omega_{i+1} = \Omega_i$ then return $\Omega_\infty = \Omega_i$, else go to 2

Solution:

- Given the feedback control law $u(k) = Kx(k)$, we obtain the autonomous system dynamics $x(k+1) = (1.5 + K)x(k)$. And the input constraint $u(k) \in \mathcal{U}$ translates to $Kx(k) \in \mathcal{U}$, resulting in constraints on the state $x(k)$:

$$
\mathcal{X} = \{x \mid Kx \in \mathcal{U}\} = \left\{ x \;\middle|\; \begin{bmatrix} Kx \\ -Kx \end{bmatrix} \leq \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} \right\}.
$$

- For a stabilizing controller we require $|(1.5 + K)| \leq 1 \Rightarrow K \in [-2.5, -0.5]$.
- 1. Step: $\Omega_0 = \mathcal{X}$
- 2. Step: Compute $\text{pre}(\mathcal{X})$

$$
\text{pre}(\mathcal{X}) = \{x | (a + bK)x \in \mathcal{X}\} = \{x | K(a + bK)x \in \mathcal{U}\} = \left\{ x \;\middle|\; \begin{bmatrix} K(1.5 + K)x \\ -K(1.5 + K)x \end{bmatrix} \leq \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} \right\}
$$

- 3. Step: Intersection (all constraints need to be satisfied):

$$
\text{pre}(\mathcal{X}) \cap \mathcal{X} = \left\{ x \;\middle|\; \begin{bmatrix} K(1.5 + K) \\ -K(1.5 + K) \\ K \\ -K \end{bmatrix} x \leq \begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} \right\} = \left\{ x \;\middle|\; \begin{bmatrix} |K(1.5 + K)x| \\ |Kx| \end{bmatrix} \leq \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} \right\}
$$

- Note that for all $K \in [-2.5, -0.5]$ it holds $|K(1.5 + K)| \leq |K|$, as we have that $|(1.5 + K)| \leq 1$. Therefore the state constraint $\mathcal{X}$ will always be more restrictive with $\mathcal{X} \subseteq \text{pre}(\mathcal{X})$ and therefore we have $\text{pre}(\mathcal{X}) \cap \mathcal{X} = \mathcal{X}$, i.e. termination of the algorithm after one step with $\Omega_\infty = \mathcal{X}$.
- The feedback K yielding the largest terminal invariant set $\mathcal{X}_f$ can therefore be found by simply maximizing the set $\mathcal{X}$, i.e., $K = -0.5$ with

$$
\mathcal{X} = \left\{ x \;\middle|\; \begin{bmatrix} -0.5x \\ 0.5x \end{bmatrix} \leq \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} \right\} = [-0.2,\ 0.2]
$$

Exercise 3

Consider the double integrator system

$$x(k+1) = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} \frac{T_s^2}{2} \\ T_s \end{bmatrix} u(k)$$

with $T_s = 0.1$ and box constraints $|u| < 0.1$, $\|x\|_\infty \leq 1$ component-wise.

a) Using the MPT toolbox (`https://www.mpt3.org/`) compute the maximum control invariant set.

b) Design a linear state feedback controller Kx and compute the maximum invariant set for the closed-loop system, resulting in an inner approximation of the control invariant set.

c) Investigate these computations for a triple integrator system

Solution:

See provided MATLAB script